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→ Exact Method

→ only applicable if solution is unique

Method of Factorization

OR

LU-Decomposition Method

Consider a system of Linear Eq

$$\left. \begin{array}{l} 3x + y - z = 3 \\ 2x - 8y + z = -5 \\ x - 2y + 9z = 8 \end{array} \right\} \text{--- (1) This system of Linear Eq can be written as}$$

$$AX = b \text{ --- (2)}$$

$$\text{let } LU = A$$

$$\text{where } L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, U = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

So ⇒ (2)

unit lower triangular

upper triangular

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

$$\begin{bmatrix} U_{11} & U_{12} & U_{13} \\ l_{21}U_{11} & l_{21}U_{12} + U_{22} & l_{21}U_{13} + U_{23} \\ l_{31}U_{11} & l_{31}U_{12} + l_{32}U_{22} & l_{31}U_{13} + l_{32}U_{23} + U_{33} \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

By Comparing

$$U_{11} = 3$$

$$U_{22} = 1$$

$$U_{33} = -1$$

$$l_{21}U_{11} = 2$$

$$l_{21}U_{12} + U_{22} = -8$$

$$l_{21}U_{13} + U_{23} = 1$$

$$l_{31}U_{11} = 1$$

$$l_{31}U_{12} + l_{32}U_{22} = -2$$

$$l_{31}U_{13} + l_{32}U_{23} + U_{33} = 9$$

→ If any value of diagonal entry of U is 0 then free variable occurs. Determinant is 0 and method not applicable

$$\Rightarrow l_{21} = 2/3$$

$$\Rightarrow l_{31} = 1/3$$

$$\Rightarrow U_{22} = -8 - 2/3 = \frac{-26}{3}$$

$$\Rightarrow U_{23} = 1 + 2/3 = 5/3$$

$$\Rightarrow l_{32} = \frac{-2 - 1/3}{-26/3} = \frac{-7/3}{-26/3} = \frac{-7}{26}$$

$$\Rightarrow U_{33} = \frac{23}{26}$$

$$\text{So, } L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 7/26 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 & -1 \\ 0 & -26/3 & 5/3 \\ 0 & 0 & 23/26 \end{bmatrix}$$

As

$$AX = b$$

$$LUX = b$$

$$(LU = A)$$

$$\text{Let } UX = Y \quad \text{--- (3)}$$

$$LY = b$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 7/26 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \\ 8 \end{bmatrix}$$

$$y_1 = 3$$

$$2/3 y_1 + y_2 = -5 \Rightarrow y_2 = -7$$

$$1/3 y_1 + 7/26 y_2 + y_3 = 8 \Rightarrow y_3 = \frac{23}{26}$$

$$\text{From (3), } UX = Y$$

$$\begin{bmatrix} 3 & 1 & -1 \\ 0 & -26/3 & 5/3 \\ 0 & 0 & 23/26 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \\ 23/26 \end{bmatrix}$$

$$3x + y - z = 3 \quad \Rightarrow x = 1$$

$$-2\frac{6}{3}y + 5\frac{1}{3}z = -7 \quad \Rightarrow y = 1$$

$$\frac{23}{26}z = \frac{23}{26} \quad \Rightarrow z = 1$$

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Cront's Method

If we take $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$, $U = \begin{bmatrix} 1 & U_{12} & U_{13} \\ 0 & 1 & U_{23} \\ 0 & 0 & 1 \end{bmatrix}$

lower triangular → Unit upper triangular

egs $\begin{cases} 3x + y - z = 3 \\ 2x - 8y + z = -5 \\ x - 2y + 9z = 8 \end{cases}$

This system of linear eq. can be written as $AX = b$

Let $LU = A$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & U_{12} & U_{13} \\ 0 & 1 & U_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

$$\begin{bmatrix} l_{11} & l_{11}U_{12} & l_{11}U_{13} \\ l_{21} & l_{21}U_{12} + l_{22} & l_{21}U_{13} + l_{22}U_{23} \\ l_{31} & l_{31}U_{12} + l_{32} & l_{31}U_{13} + l_{32}U_{23} + l_{33} \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

By Comparing

$$\boxed{l_{11} = 3}, \boxed{l_{21} = 2}, \boxed{l_{31} = 1}, \boxed{U_{12} = 1/3}, \boxed{U_{13} = -1/3}$$

$$\boxed{l_{22} = -26/3}, \boxed{U_{23} = \frac{1 - (2)(-1/3)}{-26/3} = -5/26}, \boxed{l_{32} = -2 - (1)(1/3) = -7/3}$$

$$l_{33} = 9 - (1)(-1/3) - (-7/3)(-5/26) \Rightarrow \boxed{l_{33} = \frac{231}{26}}$$

So,

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 2 & -26/3 & 0 \\ 1 & -7/3 & \frac{231}{26} \end{bmatrix}, U = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1 & -5/26 \\ 0 & 0 & 1 \end{bmatrix}$$

Let $AX=b$ then $LUX=b$ (put $U=A$)

Let $UX=Y$ then $LY=B$

$$\begin{bmatrix} 3 & 0 & 0 \\ 2 & -26/3 & 0 \\ 1 & -7/3 & 23/26 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \\ 8 \end{bmatrix}$$

So, $3y_1 = 3 \Rightarrow y_1 = 1$

$$2y_1 - \frac{26}{3}y_2 = -5 \Rightarrow y_2 = \frac{-7 \times 3}{26} = \frac{-21}{26}$$

$$y_3 = \frac{8 - y_1 + 7/3 y_2}{23/26} \Rightarrow y_3 = 1$$

From (3) $UX=Y$

$$\begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1 & -5/26 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ +21/26 \\ 1 \end{bmatrix}$$

So, $z = 1$

$$y = 1$$

$$x = 1$$

29/2/24 $\rightarrow$ Numerical method
 $\rightarrow$ solution will have error

Gauss Jacobi Method:

Ex: $8x_1 - 3x_2 + 2x_3 = 20$

$$6x_1 + 3x_2 + 12x_3 = 35$$

$$4x_1 + 11x_2 - x_3 = 33$$

Sol: To solve this system, the diagonals of A should be diagonally dominant.

$$|8| > |-3| + |2| \quad \checkmark$$

$$|3| > |6| + |12| \quad \times$$

$$|-1| > |4| + |11| \quad \times$$

which means that the given system is not diagonally dominant. By interchanging R_2 & R_3

$$|8| > |-3| + |2|, \quad |11| > |4| + |-1|, \quad |12| > |6| + |3|$$

$$8x_1 - 3x_2 + 2x_3 = 20 \quad \text{--- i)}$$

$$4x_1 + 11x_2 - x_3 = 33 \quad \text{--- ii)}$$

$$6x_1 + 3x_2 + 12x_3 = 35 \quad \text{--- iii)}$$

$$x_1 = (20 + 3x_2 - 2x_3)/8$$

$$x_2 = (33 - 4x_1 + x_3)/11$$

$$x_3 = (35 - 6x_1 - 3x_2)/12$$

1st Approximation is

$$x_1^{(1)} = (20 + 3x_2^{(0)} - 2x_3^{(0)})/8 = 2.500$$

$$x_2^{(1)} = (33 - 4x_1^{(0)} + x_3^{(0)})/11 = 3.000$$

$$x_3^{(1)} = (35 - 6x_1^{(0)} - 3x_2^{(0)})/12 = 2.917$$

$$\text{let } x_1^{(0)} = 0, \quad x_2^{(0)} = 0, \quad x_3^{(0)} = 0$$

2nd approximation is

$$x_1^{(2)} = \frac{1}{8} (20 + 3x_2^{(1)} - 2x_3^{(1)}) = 2.898$$

$$x_2^{(2)} = \frac{1}{11} (33 - 4x_1^{(1)} + x_3^{(1)}) = 2.356$$

$$x_3^{(2)} = \frac{1}{12} (35 - 6x_1^{(1)} - 3x_2^{(1)}) = 0.917$$

3rd approximation is

$$x_1^{(3)} = \frac{1}{8} (20 + 3x_2^{(2)} - 2x_3^{(2)}) = 3.154$$

$$x_2^{(3)} = \frac{1}{11} (33 - 4x_1^{(2)} + x_3^{(2)}) = 2.030$$

$$x_3^{(3)} = \frac{1}{12} (35 - 6x_1^{(2)} - 3x_2^{(2)}) = 0.880$$

4th approximation is

$$x_1^{(4)} = \frac{1}{8} (20 + 3x_2^{(3)} - 2x_3^{(3)}) = 3.041$$

$$x_2^{(4)} = \frac{1}{11} (33 - 4x_1^{(3)} + x_3^{(3)}) = 1.933$$

$$x_3^{(4)} = \frac{1}{12} (35 - 6x_1^{(3)} - 3x_2^{(3)}) = 0.832$$

5th approximation is

$$x_1^{(5)} = \frac{1}{8} (20 + 3x_2^{(4)} - 2x_3^{(4)}) = 3.017$$

$$x_2^{(5)} = \frac{1}{11} (33 - 4x_1^{(4)} + x_3^{(4)}) = 1.970$$

$$x_3^{(5)} = \frac{1}{12} (35 - 6x_1^{(4)} - 3x_2^{(4)}) = 0.913$$

6th approximation is

$$x_1^{(6)} = \frac{1}{8} (20 + 3x_2^{(5)} - 2x_3^{(5)}) = 3.011$$

$$x_2^{(6)} = \frac{1}{11} (33 - 4x_1^{(5)} + x_3^{(5)}) = 1.986$$

$$x_3^{(6)} = \frac{1}{12} (35 - 6x_1^{(5)} - 3x_2^{(5)}) = 0.916$$

7th Approximation is

$$x_1^{(7)} = \frac{1}{8} (20 + 3x_2^{(6)} - 2x_3^{(6)}) = 3.016$$

$$x_2^{(7)} = \frac{1}{11} (33 - 4x_1^{(6)} + x_3^{(6)}) = 1.988$$

$$x_3^{(7)} = \frac{1}{12} (35 - 6x_1^{(6)} - 3x_2^{(6)}) = 0.915$$

8th Approximation is

$$x_1^{(8)} = \frac{1}{8} (20 + 3x_2^{(7)} - 2x_3^{(7)}) = 3.017$$

$$x_2^{(8)} = \frac{1}{11} (33 - 4x_1^{(7)} + x_3^{(7)}) = 1.986$$

$$x_3^{(8)} = \frac{1}{12} (35 - 6x_1^{(7)} - 3x_2^{(7)}) = 0.912$$

9th Approximation is

$$x_1^{(9)} = \frac{1}{8} (20 + 3x_2^{(8)} - 2x_3^{(8)}) = 3.017$$

$$x_2^{(9)} = \frac{1}{11} (33 - 4x_1^{(8)} + x_3^{(8)}) = 1.986$$

$$x_3^{(9)} = \frac{1}{12} (35 - 6x_1^{(8)} - 3x_2^{(8)}) = 0.912$$

(56)

$$x_1 = 3.017, \quad x_2 = 1.986, \quad x_3 = 0.912$$

ہیری صورت سے بے عالم میں بہاروں کو نبات
ہیری آنکھوں کے سوا اس دنیا میں رکھا کیا ہے

Gauss Siedal Method → fastly convergent

Ex. $8x_1 - 3x_2 + 2x_3 = 20$ — i)

$4x_1 + 11x_2 - x_3 = 33$ — ii)

$6x_1 + 3x_2 + 12x_3 = 35$ — iii)

Since the diagonals of A are diagonally dominant. $\therefore$

$$x_1 = \frac{1}{8} (20 + 3x_2 - 2x_3)$$

$$x_2 = \frac{1}{11} (33 - 4x_1 + x_3)$$

$$x_3 = \frac{1}{12} (35 - 6x_1 - 3x_2)$$

1st approximation is

$$x_1^{(1)} = \frac{1}{8} (20 + 3x_2^{(0)} - 2x_3^{(0)}) = 2.500$$

$$x_2^{(1)} = \frac{1}{11} (33 - 4x_1^{(1)} + x_3^{(0)}) = 2.091$$

$$x_3^{(1)} = \frac{1}{12} (35 - 6x_1^{(1)} - 3x_2^{(1)}) = 1.144$$

2nd approximation is

$$x_1^{(2)} = \frac{1}{8} (20 + 3x_2^{(1)} - 2x_3^{(1)}) = 2.998$$

$$x_2^{(2)} = \frac{1}{11} (33 - 4x_1^{(2)} + x_3^{(1)}) = 2.019$$

$$x_3^{(2)} = \frac{1}{12} (35 - 6x_1^{(2)} - 3x_2^{(2)}) = 0.913$$

3rd approximation is

$$x_1^{(3)} = \frac{1}{8} (20 + 3x_2^{(2)} - 2x_3^{(2)}) = 3.027$$

$$x_2^{(3)} = \frac{1}{11} (33 - 4x_1^{(3)} + x_3^{(2)}) = 1.982$$

$$x_3^{(3)} = \frac{1}{12} (35 - 6x_1^{(3)} - 3x_2^{(3)}) = 0.908$$

4th Approximation is

$$x_1^{(4)} = \frac{1}{8} (20 + 3x_2^{(3)} - 2x_3^{(3)}) = 3.016$$

$$x_2^{(4)} = \frac{1}{11} (33 - 4x_1^{(4)} + x_3^{(3)}) = 1.988$$

$$x_3^{(4)} = \frac{1}{12} (35 - 6x_1^{(4)} - 3x_2^{(4)}) = 0.912$$

5th

Approximation is

$$\begin{aligned}x_1^{(5)} &= \frac{1}{8} (20 + 3x_2^{(4)} - 2x_3^{(4)}) = 3.017 \\x_2^{(5)} &= \frac{1}{11} (33 - 4x_1^{(5)} + x_3^{(4)}) = 1.986 \\x_3^{(5)} &= \frac{1}{12} (35 - 6x_1^{(5)} - 3x_2^{(5)}) = 0.912\end{aligned}$$

6th Approximation is

$$\begin{aligned}x_1^{(6)} &= \frac{1}{8} (20 + 3x_2^{(5)} - 2x_3^{(5)}) = 3.017 \\x_2^{(6)} &= \frac{1}{11} (33 - 4x_1^{(6)} + x_3^{(5)}) = 1.986 \\x_3^{(6)} &= \frac{1}{12} (35 - 6x_1^{(6)} - 3x_2^{(6)}) = 0.912\end{aligned}$$

So, $x_1 = 3.017$

$x_2 = 1.986$

$x_3 = 0.912$

1/3/24

Phase 1: Algebraic + Transcendental (Bisection, Regula Falsi, Iterative, Newton Raphson, Secant)
 Phase 2: System of Linear Eq (LU-decomposition, Gauss Jordan, Crout's, Gauss Elimination) → Give exact answer
 (Gauss Jacobi, Gauss Seidel) → Give approximate answer

Phase 3: Differential Equations.

→ Equation that contains derivative is called differential eq.

* An equation that contains the derivative of one or more dependant variable with respect to single independent variable is called O.D.E (Ordinary Differential Eq.)

eg: $\frac{dA}{dx} + \frac{dB}{dx} + C = 0$

* An equation that contains the derivative of one or more dependant variable wrt to more than one independent variable is called P.D.E (Partial Differential Eq.)

eg: $\frac{dA}{dx} + \frac{dB}{dy} + C = 0$

→ Highest derivative of equation is called Order of eq.

→ Power of highest derivative is called Degree of eq.

→ Solution of D.E $\frac{dy}{dx} = f(x, y)$ will be in the form $y = \dots$

1. ODE 2. PDE
Hmara level

3. Directional derivative
4. co-variant derivatives
5. Lee derivatives
↳ sir Anjod ka level

EULER'S METHOD start of Problem

Consider the ODE $\frac{dy}{dx} = f(x, y)$; $y(x_0) = y_0$

Suppose that we wish to find

$y_1, y_2, y_3, \dots, y_m$ where y_m is the value of

y corresponding to x_m

$$y - y_0 = \left(\frac{dy}{dx} \right)_{x=x_0} (x - x_0)$$

$$y_1 = y_0 + (x_1 - x_0) f(x_0, y_0)$$

$$y_1 = y_0 + h f(x_0, y_0)$$

$$y_2 = y_1 + h f(x_1, y_1)$$

$$y_3 = y_2 + h f(x_2, y_2)$$

$\vdots$

$$y_{m+1} = y_m + h f(x_m, y_m)$$

$$h = \frac{b-a}{n}$$

where b is the final point, a is initial point, n is number of steps, h is step size

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Ex: $\frac{dy}{dx} = f(x, y)$ $dy = (1-y)dx$

$\frac{dy}{dx} = 1-y$; $y(0) = 0$; $0 \leq x \leq 0.3$

$\frac{dy}{1-y} = dx \Rightarrow \int \frac{dy}{1-y} = \int dx$

$\Rightarrow -\ln|1-y| = x + \ln|c|$

$\ln\left|\frac{c}{1-y}\right| = x \Rightarrow 1-y(e^x) = c$

put IVP.

$c=1$. So, $y = 1 - e^{-x}$

$x_0 = 0$

$x_1 = x_0 + h = 0 + 0.1 = 0.1$

$x_2 = x_1 + h = 0.2$

$x_3 = x_2 + h = 0.3$

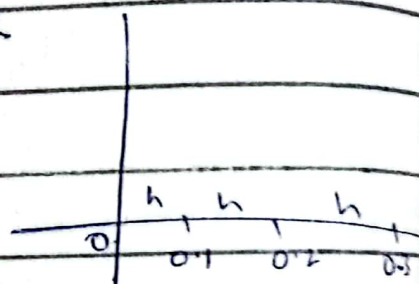
$y(x) = 1 - e^{-x}$

$y(x_0) = 1 - e^0 = 0 = y(0)$

$y(x_1) = y(0.1) = 0.0951625$

$y(x_2) = y(0.2) = 0.1812692$

$y(x_3) = y(0.3) = 0.2591817$



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EULER'S METHOD

eg: $\frac{dy}{dx} = 1-y$; $y(0)=0$; $0 \leq x \leq 0.3$

Sol: $y_{m+1} = y_m + h f(x_m, y_m)$

let $h = 0.1$

$f(x, y) = 1-y \Rightarrow f(x_m, y_m) = 1-y_m$

So, $y_{m+1} = y_m + h(1-y_m)$

$y_{m+1} = h + (1-h)y_m$

$y_{m+1} = 0.1 + 0.9y_m$

now put $m=0$

$y_1 = 0.1 + 0.9(0)$

$y_1 = 0.1$

$x_0 = 0$

$x_1 = x_0 + h = 0.1$

$x_2 = x_1 + h = 0.2$

$x_3 = x_2 + h = 0.3$

$y_0 = y(x_0) = 0$

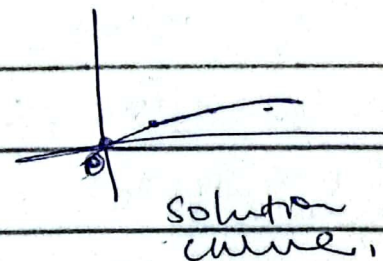
$y_2 = 0.1 + 0.9y_1 = 0.1 + 0.9(0.1) = 0.19$

$y_3 = 0.1 + 0.9y_2 = 0.1 + 0.9(0.19) = 0.271$

So,

$(0, 0), (0.1, 0.1), (0.2, 0.19), (0.3, 0.271)$

is the solution of ODE



ex. $\frac{dy}{dx} = 1-y$; $y(0)=0$; $0 \leq x \leq 0.3$

$$y_{m+1} = y_m + h f(x_m, y_m)$$

let $h = 0.05$

So, $x_0 = 0$

$$x_1 = x_0 + h = 0.05$$

$$x_2 = x_1 + h = 0.10$$

$$x_3 = x_2 + h = 0.15$$

$$x_4 = x_3 + h = 0.20$$

$$x_5 = x_4 + h = 0.25$$

$$x_6 = x_5 + h = 0.30$$

So, $y_{m+1} = 0.05 + 0.95 y_m$

$$y_0 = y(x_0) = 0$$

$$y_1 = y(x_1) = 0.05 + 0.95(0) = 0.05$$

$$y_2 = y(x_2) = 0.05 + 0.95(0.05) = 0.10$$

$$y_3 = y(x_3) = 0.05 + 0.95(0.10) = 0.15$$

$$y_4 = y(x_4) = 0.05 + 0.95(0.15) = 0.19$$

$$y_5 = y(x_5) = 0.05 + 0.95(0.19) = 0.23$$

$$y_6 = y(x_6) = 0.05 + 0.95(0.23) = 0.26$$

<u>Table:</u> x	y	Exact sol	Euler Method	Error
x_0	y_0	$y_0 =$	$y_0 =$	Exact - Euler
x_1	y_1	$y_1 =$	$y_1 =$	
x_2	y_2	$y_2 =$	$y_2 =$	
x_3	y_3	$y_3 =$	$y_3 =$	

eg. $\frac{dy}{dx} = y - \frac{2x}{y}$; $y(0) = 1$; $0 \leq x \leq 2$

let $h = 0.4$

$x_0 = 0$

$x_1 = x_0 + h = 0.4$

$x_2 = x_1 + h = 0.8$

$x_3 = x_2 + h = 1.2$

$x_4 = x_3 + h = 1.6$

$x_5 = x_4 + h = 2.0$

So, $y_0 = y(x_0) = 1$ (Initial condition)

$y_1 = y_0 + h \left(y_0 - \frac{2x_0}{y_0} \right) = 1 + 0.4 \left(1 - \frac{0}{1} \right) = 1.4000$

$y_2 = y_1 + h \left(y_1 - \frac{2x_1}{y_1} \right) = 1.4 + 0.4 \left(1.4 - \frac{2(0.4)}{1.4} \right) = 1.7314$

$y_3 = 2.0543$

$y_4 = y_3 + h \left(y_3 - \frac{2x_3}{y_3} \right) = 2.0543 + 0.4 \left(2.0543 - \frac{2(1.2)}{2.0543} \right) = 2.4087$

$y_5 = 2.8908$

$y_6 = y_4 + h \left(y_4 - \frac{2x_4}{y_4} \right) = 2.8908$

for h :

$h = \frac{b-a}{n}$

interval $\rightarrow$ end point $\rightarrow$ start point $\rightarrow$ number of steps

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Improved EULER'S METHOD

$$If \quad \frac{dy}{dx} = f(x, y) \quad \text{and} \quad y(x_0) = y_0$$

$$y_{m+1} = y_m + \frac{h}{2} \left[f(x_m, y_m) + f(x_m + h, y_m + h f(x_m, y_m)) \right]$$

eg. $\frac{dy}{dx} = y - \frac{2x}{y} \quad ; y(0) = 1 \quad ; 0 \leq x \leq 0.2$

let $h = 0.1 \quad x_0 = 0 \quad , x_1 = x_0 + h = 0.1, x_2 = x_1 + h = 0.2$

$$f(x_m, y_m) = y_m - \frac{2x_m}{y_m}$$

$$y_{m+1} = y_m + \frac{h}{2} \left[y_m - \frac{2x_m}{y_m} + f(x_m + h, y_m + h(y_m - \frac{2x_m}{y_m})) \right]$$

$$y_{m+1} = y_m + \frac{h}{2} \left[y_m - \frac{2x_m}{y_m} + y_m + h(y_m - \frac{2x_m}{y_m}) - \frac{2(x_m + h)}{y_m + h(y_m - \frac{2x_m}{y_m})} \right]$$

put $m = 0$

$$y_1 = y_0 + \frac{h}{2} \left[y_0 - \frac{2x_0}{y_0} + y_0 + h(y_0 - \frac{2x_0}{y_0}) - \frac{2(x_0 + h)}{y_0 + h(y_0 - \frac{2x_0}{y_0})} \right]$$

$$y_1 = 1.09591$$

$$y_2 = y_1 + \frac{h}{2} \left[f(x_1, y_1) + f(x_1 + h, y_1 + h f(x_1, y_1)) \right]$$

$$= 1.09591 + \frac{0.1}{2} \left[f(0.1, 1.09591) + f(0.2, 1.09591 + 0.1 f(0.1, 1.09591)) \right]$$

$$= 1.09591 + \frac{0.1}{2} \left[0.91341 + f(0.2, 1.18725) \right]$$

باقی کا جواب

15/8/24

MODIFIED EULER'S METHOD

$$\text{If } \frac{dy}{dx} = f(x, y) ; y(x_0) = y_0 ; x_0 \leq x \leq b$$

then

$$y_{m+1} = y_m + hf\left(x_m + \frac{h}{2}, y_m + \frac{h}{2}f(x_m, y_m)\right)$$

ex. $\frac{dy}{dx} = y - \frac{2x}{y} ; y(0) = 1 ; 0 \leq x \leq 0.2$

$$h = \frac{b-a}{4} = \frac{0.2-0}{4}$$

$$x_0 = 0$$

$$y_0 = y(x_0) = 1$$

$$h = 0.05$$

$$x_1 = x_0 + h = 0 + 0.05 = 0.05$$

$$x_2 = x_1 + h = 0.10$$

$$x_3 = x_2 + h = 0.15$$

$$x_4 = x_3 + h = 0.20$$

Sol

$$y_1 = y_0 + hf\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}f(x_0, y_0)\right)$$

$$= 1 + 0.05 f(0.025, 1.0250)$$

$$y_1 = 1 + 0.05 \left(\frac{0.0250 - 2(0.025)}{1.0250} \right)$$

$$y_1 = 1.04881$$

$$y_2 = y_1 + hf\left(x_1 + \frac{h}{2}, y_1 + \frac{h}{2}f(x_1, y_1)\right) = 1.04881 + 0.05 f(0.0750, 1.07265)$$

$$y_2 = 1.04881 + 0.05 f(0.0750, 1.07265)$$

$$y_2 = 0.94921$$

22/3/24 → 4th Order
Runge Kutta Method (RK method)

9. $\frac{dy}{dx} = f(x, y) ; y(x_0) = y_0 ; x_0 \leq x \leq b$

So, $y_1 = y_0 + k$

where $k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$

when

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

Also, $y_2 = y_1 + k$

where $k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$

$$k_1 = hf(x_1, y_1)$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_1 + h, y_1 + k_3)$$

⋮

And so on.

eg: $y' = x^2 - y$; $y(0) = 1$; $h = 0.1$

$x_0 = 0$; $x_1 = 0.1$; $x_2 = 0.2$; $x_3 = 0.3$; ...

$y_1 = y_0 + k$

$k_1 = 0.1 f(x_0, y_0) = 0.1 (0 - 1) = -0.1$

$k_2 = h f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}) = 0.1 (0.0500, 0.9500) = 0.1 (-0.9475)$

$k_2 = -0.09475$

$k_3 = h f(x_0 + h, y_0 + k_1) = 0.1 (0.1000, 0.9049) = -0.09501$

$k_4 = h f(x_0 + h, y_0 + k_1) = 0.1 (0.1, 0.9049) = -0.08950$

$k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] = -0.09464$

So, $y_1 = y_0 + k = 1 - 0.06321 = 0.93679$

Now, $y_2 = y_1 + k$

$k_1 = h f(x_1, y_1) = 0.1 (0.1, 0.93679) = -0.08452$

$k_2 = h f(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}) = 0.1 (0.1500, 0.86619) = 0.1 (-0.83701)$

$k_2 = -0.08370$

$k_3 = h f(x_1 + h, y_1 + k_1) = 0.1 (0.2000, 0.85227) = -0.08408$

$k_4 = h f(x_1 + h, y_1 + k_1) = 0.1 (0.2, 0.82108) = -0.07811$

$k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] = -0.08390$

$y_2 = y_1 + k = 0.82126$

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RK Method^{of order} to solve System of ODE

→ Collection of one or more differential equations is called System of ODE.

* Consider the following system of ODE

$$\frac{dy}{dx} = f_1(x, y, z) \quad ; \quad y(x_0) = y_0$$

$$\frac{dz}{dx} = f_2(x, y, z) \quad ; \quad z(x_0) = z_0$$

Now starting from (x_0, y_0, z_0) , The increment in k & l in y & z respectively are given by

$$y_1 = y_0 + k \quad ; \quad z_1 = z_0 + l$$

$$k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$l = \frac{1}{6} [l_1 + 2l_2 + 2l_3 + l_4]$$

$$k_1 = hf_1(x_0, y_0, z_0)$$

$$l_1 = hf_2(x_0, y_0, z_0)$$

$$k_2 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) \quad ; \quad l_2 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right)$$

$$k_3 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) \quad ; \quad l_3 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right)$$

$$k_4 = hf_1(x_0 + h, y_0 + k_3, z_0 + l_3) \quad ; \quad l_4 = hf_2(x_0 + h, y_0 + k_3, z_0 + l_3)$$

Next

$$y_2 = y_1 + k$$

Next

$$z_2 = z_1 + l$$

eg =

$$dy/dx = yz + x \quad ; \quad y(0) = 1$$

$$dz/dx = xz + y \quad ; \quad z(0) = -1$$

$$h = 0.1$$

$$y_1 = y_0 + k$$

;

$$z_1 = z_0 + l$$

$$k_1 = h f_1(0, 1, -1)$$

$$l_1 = h f_2(0, 1, -1)$$

$$= h(-1 + 0)$$

$$= h(0 + 1) = h(0 + 1)$$

$$= -0.1$$

$$= 0.1$$

$$k_2 = h f_1(0.05, 0.95, -0.95)$$

$$l_2 = h f_2(0.05, 0.95, -0.95)$$

$$= 0.1(-0.90250 + 0.05)$$

$$= 0.1(-0.04750 + 0.95)$$

$$= 0.1(-0.85250)$$

$$= 0.1(0.90250)$$

$$= -0.08525$$

$$= 0.09025$$

$$k_3 = h f_1(0.05, 0.95738, -0.95488)$$

$$l_3 = h f_2(0.05, 0.95738, -0.95488)$$

$$= 0.1(-0.91418 + 0.05)$$

$$= 0.1(0.90564)$$

$$= -0.08642$$

$$= 0.09096$$

$$k_4 = h f_1(0.1, 0.91358, -0.90904)$$

$$l_4 = h f_2(0.1, 0.91358, -0.90904)$$

$$= 0.1(-0.83047 + 0.1)$$

$$= 0.1(-0.09090 + 0.91358)$$

$$= -0.07305$$

$$= 0.08227$$

$$k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$l = \frac{1}{6} [l_1 + 2l_2 + 2l_3 + l_4]$$

$$k = -0.08607$$

$$= 0.09078$$

$$y_1 = 1 + (-0.08607) = 0.91393$$

$$z_1 = -1 + 0.09078 = -0.90922$$

Genuine

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لا حول ولا قوة الا بالله

RK Method to Solve Higher Order ODE:

Example 3

$$\frac{d^2 y}{dx^2} = x \frac{dy}{dx} - y \quad ; \quad y(0) = 3 \quad ; \quad y'(0) = 0$$

$$y = z$$

$$z' = y' = w$$

$$w' = z'' = y'' = xy' - y$$

So,

$$\frac{dz}{dx} = w$$

$$\frac{dw}{dx} = xw - z$$

$$\left. \begin{array}{l} z(0) = 3 \\ w(0) = 0 \end{array} \right\}$$

$$\text{let } h = 0.1$$

$$z_1 = z_0 + k_1 = 2.98499$$

$$k_1 = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] = -0.01501$$

$$k_1 = hf_1(x_0, z_0, w_0) = 0$$

$$k_2 = hf_1\left(x_0 + \frac{h}{2}, z_0 + \frac{k_1}{2}, w_0 + \frac{l_1}{2}\right) = -0.015$$

$$k_3 = hf_1\left(x_0 + \frac{h}{2}, z_0 + \frac{k_2}{2}, w_0 + \frac{l_2}{2}\right) = -0.01504$$

$$k_4 = hf_1(x_0 + h, z_0 + k_3, w_0 + l_3) = -0.03$$

Also,

$$w_1 = w_0 + l_1 = -0.30050$$

$$l_1 = hf_2(x_0, z_0, w_0) = -0.3$$

$$l_2 = hf_2\left(x_0 + \frac{h}{2}, z_0 + \frac{k_1}{2}, w_0 + \frac{l_1}{2}\right) = -0.30075$$

$$l_3 = hf_2\left(x_0 + \frac{h}{2}, z_0 + \frac{k_2}{2}, w_0 + \frac{l_2}{2}\right) = -0.3$$

$$l_4 = hf_2(x_0 + h, z_0 + k_3, w_0 + l_3) = -0.30130$$

$$l = -0.30050$$

$$\frac{d^2 y}{dx^2} = xy'' - y' - y$$

$$\text{let } y = z$$

$$z' = y' = \phi$$

$$\phi' = y'' = z'' = \psi$$

$$\psi' = y''' = z''' = x\psi - \phi - z$$

So,

$$\frac{dz}{dx} = \phi$$

$$\frac{d\phi}{dx} = \psi$$

$$\frac{d\psi}{dx} = x\psi - \phi - z$$

for 3rd order ODE

n -order ODE can be converted to n 1st order
system of ODE

Regards ,

